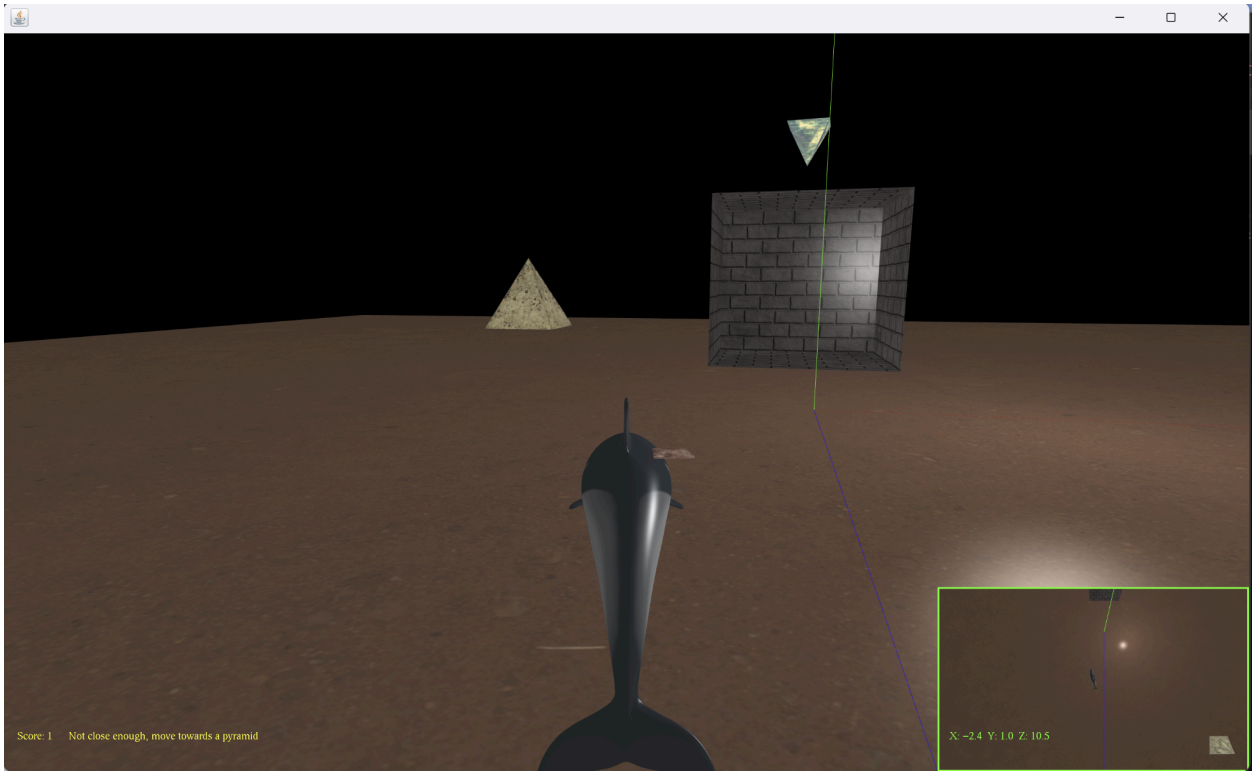
1. Tatiana Neville, CSC-165-02, A2 – Dolphin Archeologist – Spring 2026

2. a screenshot showing a typical scene from game



In this game you, a Dolphin, move around the world in search of pyramids to photograph. These pyramids are a bit shy, and once their picture is taken they move away! Collect pictures of all 3 scattered pyramids and return to your home to put your new found treasures on the wall.

3. a list of keyboard and gamepad controls for moving the dolphin

- Keyboard Keybinds
  - **Dolphin Movement**
    - *W:* While pressed moves dolphin forwards
    - *S:* While pressed moves dolphin backwards
    - *A:* While pressed moves dolphin to the left, Yaw left
    - *D:* While pressed moves dolphin to the right, Yaw right

  - **Game Controls**
    - *P:* Takes a picture of pyramid when close enough
    - *Space:* After all pictures taken and Dolphin avatar is in home, pressing space will put the pictures taken onto the back wall
    - *X:* Toggles visibility of colored XYZ axis markers

- Gamepad keybinds

    - Forwards, Backwards, Yaw left and Yaw Right mapped to the Right joystick of XBOX controller

4. a list of keyboard and gamepad controls for the orbit camera and overhead camera

- Keyboard Keybinds
    - **Orbit Camera Controls**
        - *Up Arrow:* While pressed moves the orbit camera elevation up
        - *Down Arrow:* While pressed moves orbit camera elevation down
        - *Left Arrow:* While pressed moves orbit camera azimuth left
        - *Right arrow:* While pressed moves orbit camera azimuth right
        - *Q:* While pressed moves orbit camera radius in
        - *E:* While pressed moves orbit camera radius out

    - **Overhead Camera Controls**
        - *I:* While pressed pans overhead camera up
        - *K:* While pressed pans overhead camera down
        - *J:* While pressed pans overhead camera left
        - *L:* While pressed pans overhead camera right
        - *O:* While pressed zooms overhead camera in
        - *U:* While pressed zooms overhead camera out
        - *H:* When pressed centers overhead camera on avatar and locks view to follow avatar, panning overhead camera will unlock following avatar POV

- Gamepad keybinds
    - None

5. a description of your node controllers and what they do, and how they are used in your game
- **RotationController Node:** This node rotates a game object around a given axis, <u>this was given in TAGE.</u> I used this to rotate a marker above the Dolphin's home to mark it as an important location.
- **ShakeSinkController Node:** This node shakes a game object in the X and Z direction while slowly sinking down its Y axis. I used this controller as an effect on the pyramids after they had their picture taken. I wanted to use this as an effect to eliminate any pyramids that had their picture already taken.

6. a list of all changes made to the TAGE engine
- **GameObject Changes**
    - **yaw:** rotates GameObject left/right around its current world up axis, pre-multiplies and updates localRotation and world transforms
    - **globalYaw:** Rotates GameObject left/right around the global Y axis
    - **pitch:** Rotates GameObject up/down around its current world right axis, pre-multiplies and updates localRotation and world transforms

- **Camera Changes**
    - **yaw:** Rotates the camera left/right around its local V axis and updates the U and N vectors while keeping V fixed
    - **pitch:** Rotates the camera up/down around its local U axis and updates the V and N vectors while keeping U fixed

- **CameraOrbit3D Creation**
    - CameraOrbit3D is a 3rd-person camera controller that causes a camera to orbit around a target avatar at a given azimuth, elevation, and radius
    - Supports orbiting left/right, adjusting elevation, and zooming in/out but adjusting radius

- **ShakeSinkController Creation**
    - ShakeSinkController is a node controller that causes an object to shake in place and slowly sink downwards, this gives the object a spiral appearance
    - Supports adjustable shake speed, adjustable shake intensity, and adjustable sinking speed. Uses elapsed time for consistent sinking.

- **ManualPyramid Creation**
    - ManualPyramid is a manual object shaped as a pyramid, <u>this code was given to us as an example of a manual object in our class.</u>

7. a list of any requirements that you weren't able to get working
- **Following requirements fulfilled:**
    - Fully functional orbit camera attached to dolphin avatar
    - second viewport with fully functional overhead camera
    - HUDs for both viewports that reposition automatically with resizing of window
    - ground plane at y=0, appropriate status messages displayed based on game progress
    - two different node controllers implemented (rotation controller given, shake sink controller created)
    - each pyramid textured differently (2 textures from the web, 1 self made)
    - hierarchical relationship between two or more GameObjects (photos and dolphin, photos and home, home marker and home),
    - yaw changed to globalYaw
    - hop on/off dolphin removed and space bare only transfers photos to wall of home
- **No assignment requirements left unfulfilled.**

8. Assets used in game
- Models
    - Dolphin - Given with Dolphin Assignment
- Textures
    - Dolphin_HighPolyUV.jpg - Given with Dolphin Assignment
    - Brick1.jpg - Given with TAGE

    - Ice.jpg - Given with TAGE
    - Tatertots.jpg - Custom made by Me
    - Makemake.jpg - Solar System Scope (https://www.solarsystemscope.com/textures/) License at bottom of page (https://creativecommons.org/licenses/by/4.0/)
    - Eris.jpg - Solar System Scope (https://www.solarsystemscope.com/textures/) License at bottom of page (https://creativecommons.org/licenses/by/4.0/)
    - Sand.jpg - Polyhaven (https://polyhaven.com/a/red_sand) License (https://polyhaven.com/license)
- Code
    - ManualPyramid - Given in Class Example

9. Tested on lab machine **ECS-MYST**